



One for All: RVCK & BootSTD on SBCs

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- BootSTD

“Standard Boot” implemented by U-Boot

Replacing existing DistroBoot scripts

<https://docs.u-boot.org/en/latest/develop/bootstd/overview.html>

- RVCK

RISC-V Common Kernel

Serves as the kernel source of openEuler RISC-V

Focus on RISC-V enablement

- Joined OERV in 2024 H2
- Responsibilities

Tracking the progress of RISC-V Server Platform

- Backport mainline Linux features into RVCK
- Tracking the progress of Boot Components

Make OERV available on embeddeds (particularly, SBCs)

- Shipped with vendor kernel, but with OERV kernel features applied.
- Let RVCK support those SoCs as well.

All in One

- OERV releases:
 - Pre-built full-disk image file
 - Tools to reliably, ****safely**** re-create (and customize) the image
- Boards/SoCs boot components
 - Clear understanding of what BootROM can do (per SoC)
 - Reproducible (and customizable) boot components
 - [BONUS] Reuse boot components with other distros
- Documents and guidance: <https://images.oerv.ac.cn/>
- Compatibility with server platform

- Boot component architecture:
 - Reduced BootROM + U-Boot SPL + OpenSBI + U-Boot Proper
 - Complex BootROM / proprietary FSBL + OpenSBI + U-Boot
 - Uncommon ones: EDK2, LinuxBoot, ...
- Peripheral firmware
 - Kernel Direct Filesystem Lookup

- Still relies on old distro-boot scripts
 - With fixed filesystem UUID, limited boot flow
- Some boards leave the NOR flash unsoldered!
 - But almost all SoCs supports booting from SPI flash.
- Heavily customized, breaking all standard usage
 - :(
- Good News: RISC-V eco is young enough!

- SPI NOR flash served as firmware storage
Well established solution, for binding fw with particular hw.
- Deploy U-Boot Standard Boot
Well defined generic boot logic
- Make use of extlinux.conf as boot entry provider
The only one that supports runtime decision of FDT source
May be obsoleted in the future, but will never break
- Reuse U-Boot among boards with same SoC

- Roadmap: U-Boot UEFI + FDT + GRUB2
- Why UEFI?
 - ISO installer only supports that
- Why not ACPI?
 - Eventually becoming yet another FDT on hw fragments
 - Even worse, tying firmware and kernel altogether
- Any progress?
 - K1 verification

- 两个和尚抬水吃

U-Boot can not decide FDT from which kernel to use

- It need to tell GRUB2 about hw info as well

GRUB2 can not decide FDT for which hardware to use

- FDT support is relatively new in GRUB2, with no existing boot scripts

- RVCK provides kernel support for hardwares
- Unified ISO installer depends on RVCK
- Standard boot components support operation of ISO installer, as well as its installation.
 - UEFI + ACPI on Server Platform
 - UEFI + FDT on embeddeds
- => One for All